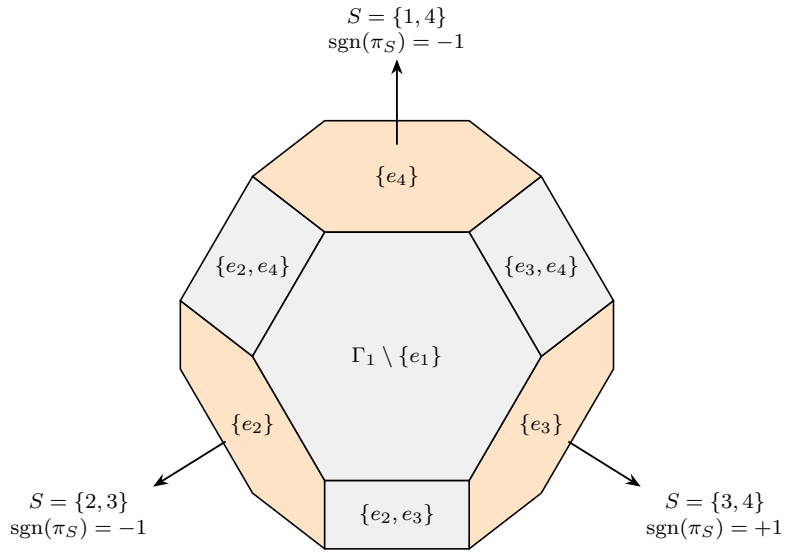
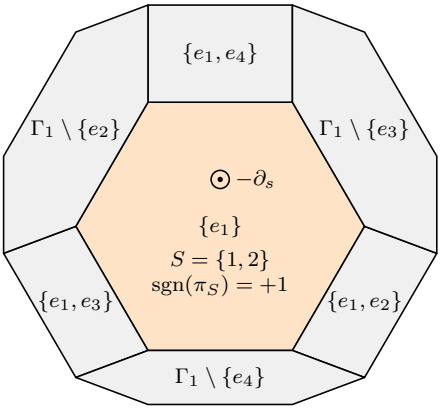
(a) $(\widehat{0,+\infty})^4/\mathbb{R}_+$ for the square ring, front

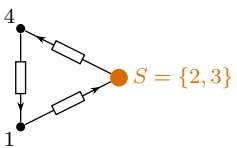


(b) back



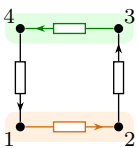
(c) The 14 faces, $\emptyset \neq \gamma \subsetneq \Gamma_1$

Laman island



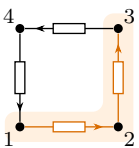
$\{e_j\}$, $\mathrm{sgn}(\pi_S) = (-1)^{j-1}$

two islands



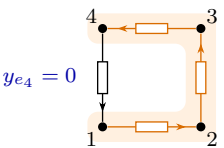
$\{e_1, e_3\}$, $\{e_2, e_4\}$

too few resistors



$\{e_1, e_2\}$, $\{e_2, e_3\}$, $\{e_3, e_4\}$, $\{e_1, e_4\}$
$|\Gamma_1[S]| = 2 < 2|S| - 3$

bypassed resistor



$\Gamma_1 \setminus \{e_j\}$

$$0 = \sum_{\emptyset \neq \gamma \subsetneq \Gamma_1} \int_{\{s=0\}} e^{-\sum_e z_e^{+\dot{a}} y_{e,+\dot{a}}} \bigwedge_{e \notin T} \left(-d^2 y_e\right) = \sum_S \mathrm{sgn}(\pi_S)\, \mathcal{I}^{\text{raw cut}}_{\Gamma[S]}(\lambda + \partial; z)\, \mathcal{I}^{\text{raw cut}}_{\Gamma(S)}(\lambda; z)$$

$$= \sum_{j=1}^{4} (-1)^{j-1}\, \mathcal{I}^{\text{raw cut}}_{e_j}(\lambda + \partial; z_{e_j})\, \mathcal{I}^{\text{raw cut}}_{\Gamma/e_j}(\lambda; z)$$